Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

PROJECT TITLE :-

" To reduce cost of cotton cloth usage in STCF by 40%"

Mahindra and Mahindra Ltd, Nasik Plant, Scorpio TCF.

	Plan	Actual
Start Date	**04/01/2019**	**5/2/2019**
Completion Date	**31/12/2019**	**31/12/2019**

Current Status: **Closed**

Cross Functional Team

Sr. No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Nitin Aher	L6-M	STCF	9960800220
2	**Mentor**	Rahul Narkar	DGM	STCF	9881080299
2	**Member 1**	Pravin Patil	L7-M	STCF	8888897850
3	**Member 2**	Niranjan Jadhav	L7-O	STCF	8055709002
4	**Member 3**	Naziya Bhagwan	L8-O	SCM	8600828329
5	**Member 4**	Amit Patwardhan	L7-O	PAINT PU	9881147491
6	**Member 5**	Mahesh Savkar	L7-O	STCF	8888897833

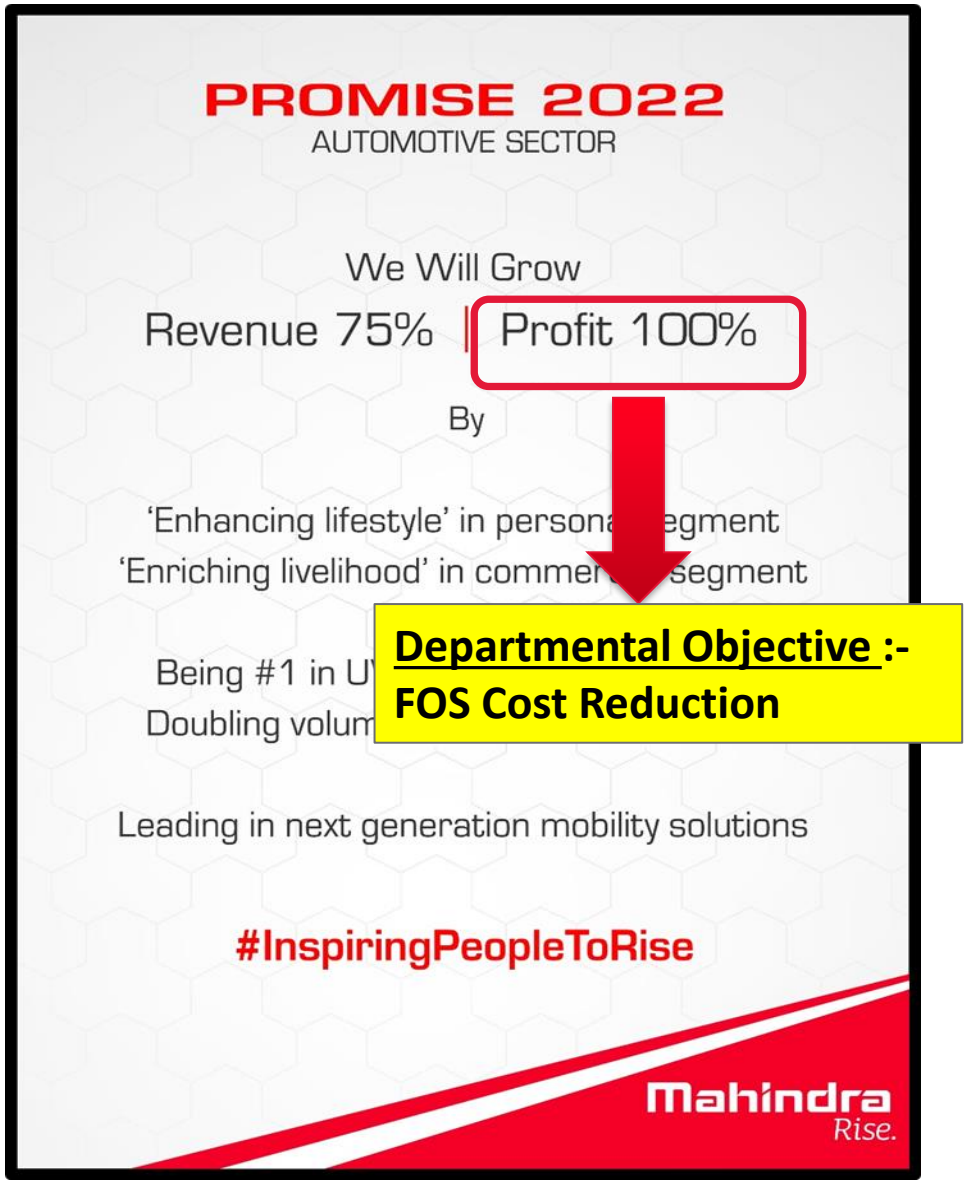
Step1. Select Subject

Brief Description : Excess consumption of cotton cloth in STCF CAI since F18 , avg 100 kg /month .

- What : Excess consumption of cotton in STCF .
- Where : At CAI During Vehicle cleaning in wrap & paint booth.
- When : Since F18.
- Who : Associates on vehicle cleaning & Paint touch up stages.
- Why : FOS cost fired.
- How : 100 kg / month usage of cotton for activity.

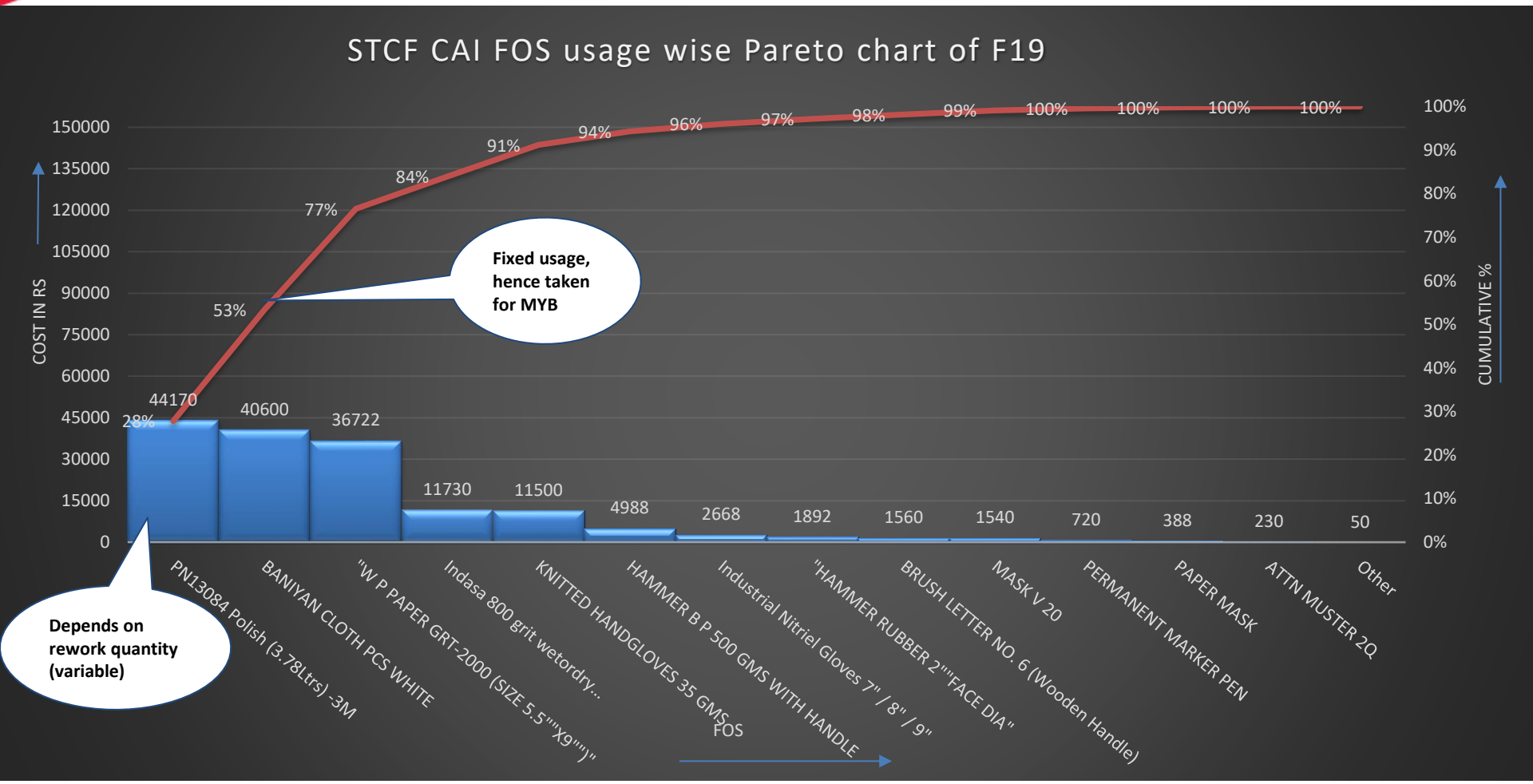
Cotton waste after usage.

2. Justify the choice (reasons)



Departmental Objective :-
FOS Cost Reduction

2. Justify the choice



INFERENCE :-

AS PER PARETO GRAPH BANIYAN CLOTH IS THE TOP **FIXED** USAGE IN FOS

2. Justify the choice

Consumption of cloth month wise in KG

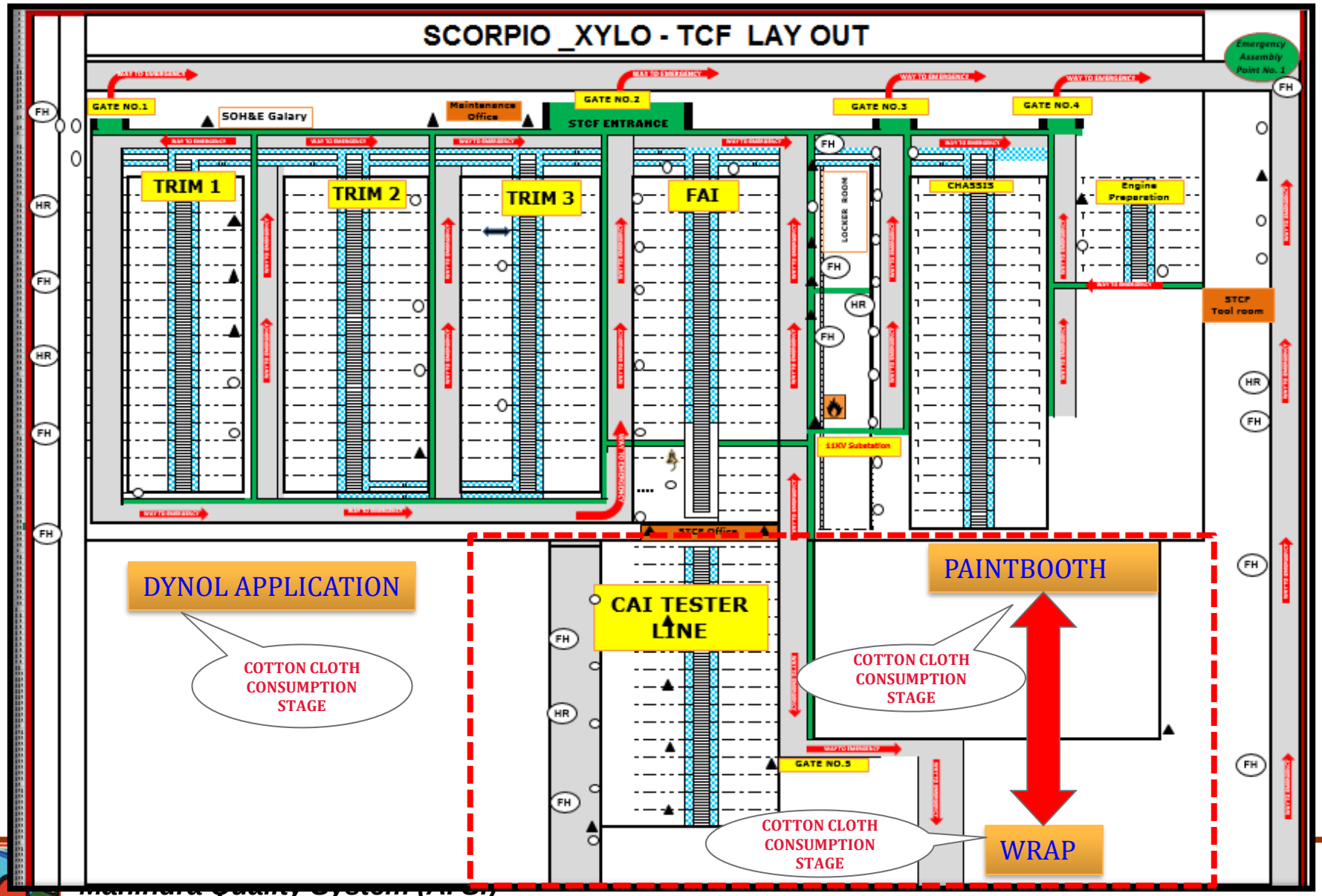
Month	F19
Apr	100
May	110
Jun	90
Jul	175
Aug	215
Sep	150
Oct	160
Nov	105
Dec	100
Total	1215

F19 RS : 70760 Cost for cloth .

2.b Immediate Containment Action

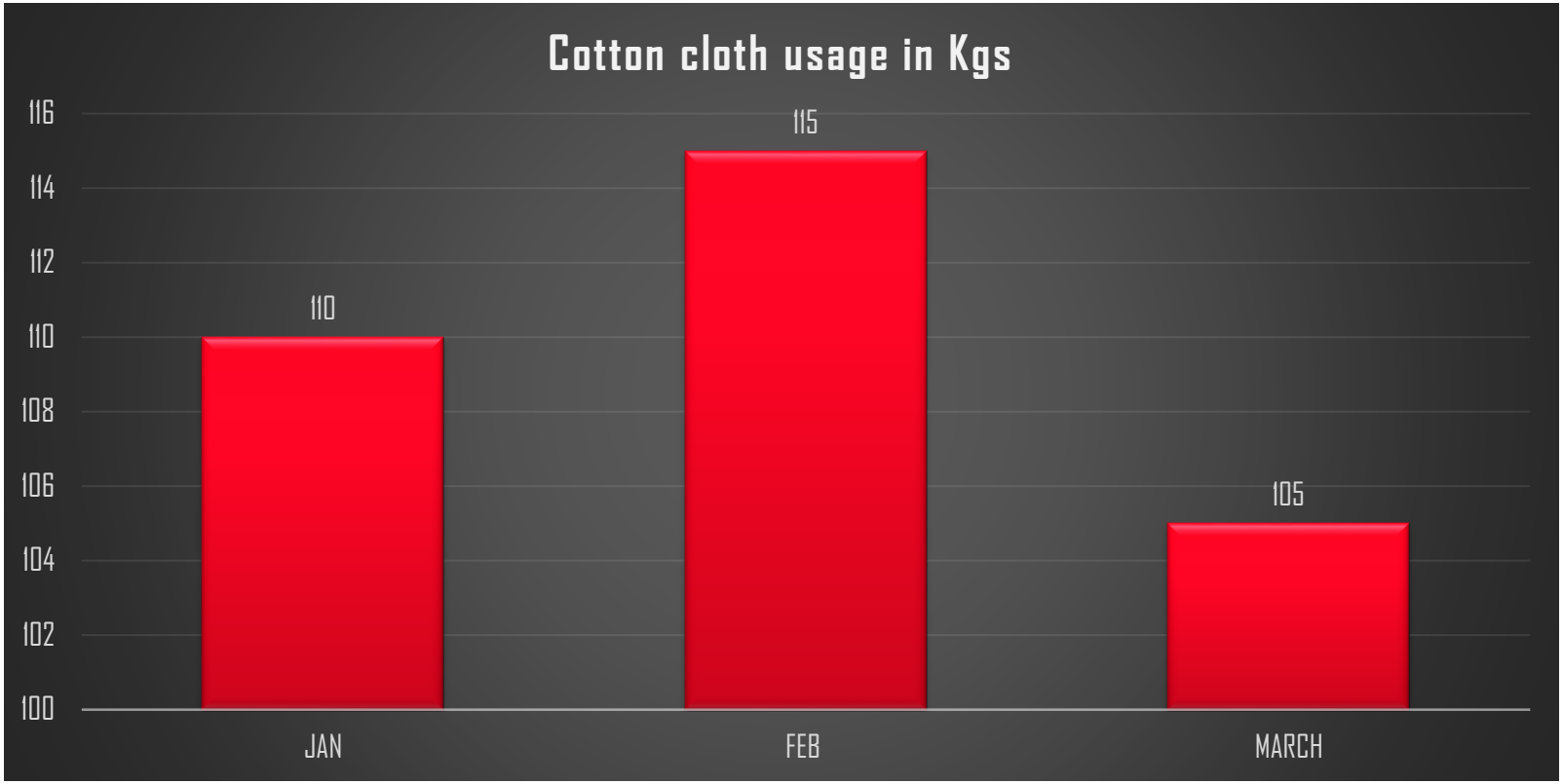
- ❑ **Control cotton cloth issuance started with permission of manager.**
- ❑ **Lock and Key arrangement made for cotton cloth storage in shop floor.**

Step 3. Understanding Current Situation



3. Understand the current situation

Month wise Trend



INFERENCE :- As per graph cotton usage is consistent month on month

Step 3. Understanding Current Situation

Good & Not so good observations

- No Monitoring of cotton cloth usage.
- No fix size pieces of cotton cloth available.
- Large size cotton cloth used and thrown .

Step 3. Understanding Current Situation

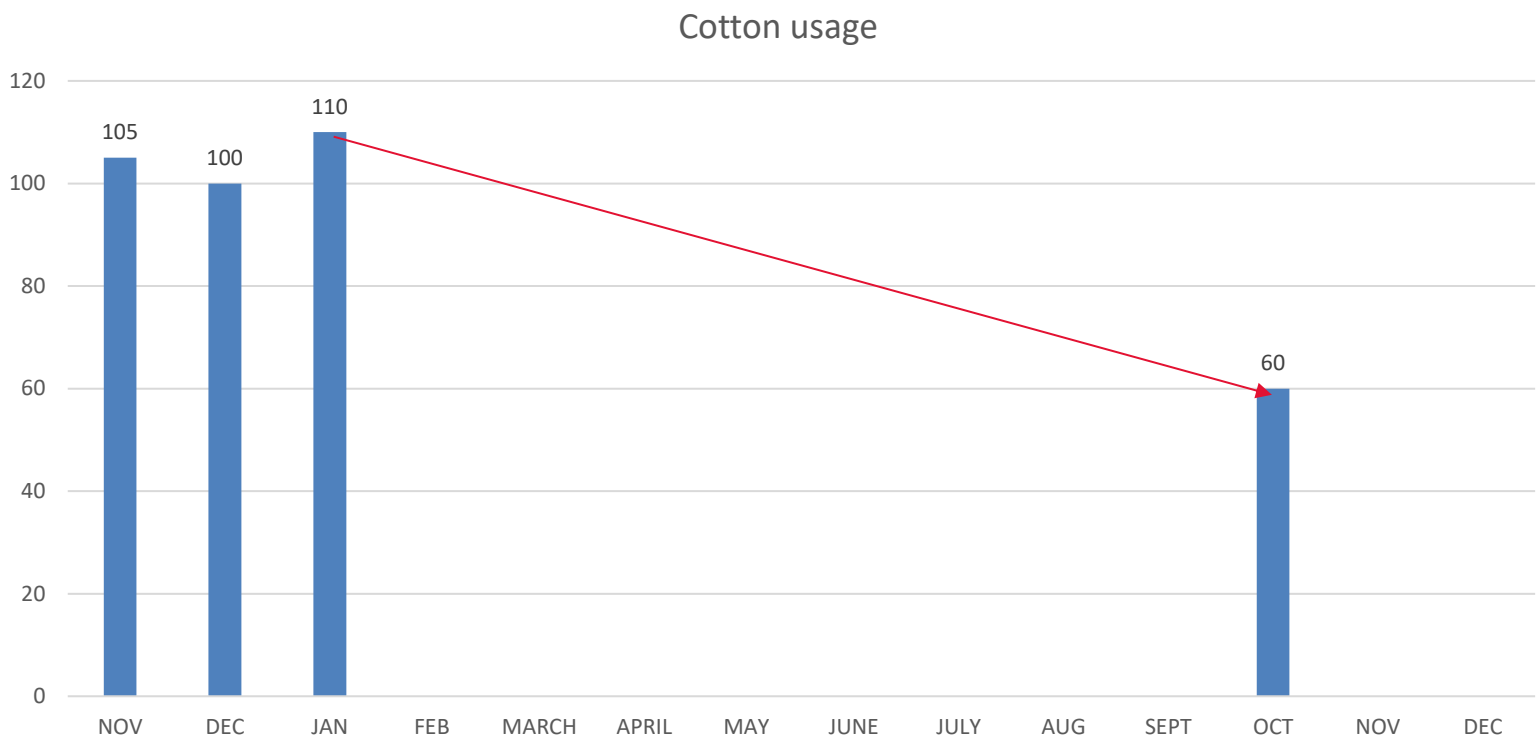
Areas Involved in the analysis:

1. Paint Booth.
2. Wrap Vehicle Cleaning.
3. Dynol cleaning.

4. Select targets

Present Level: 315 kg cotton cloth used at CAI in last 3 months.

Target : **40 % Reduction by Oct 2019**



TARGET

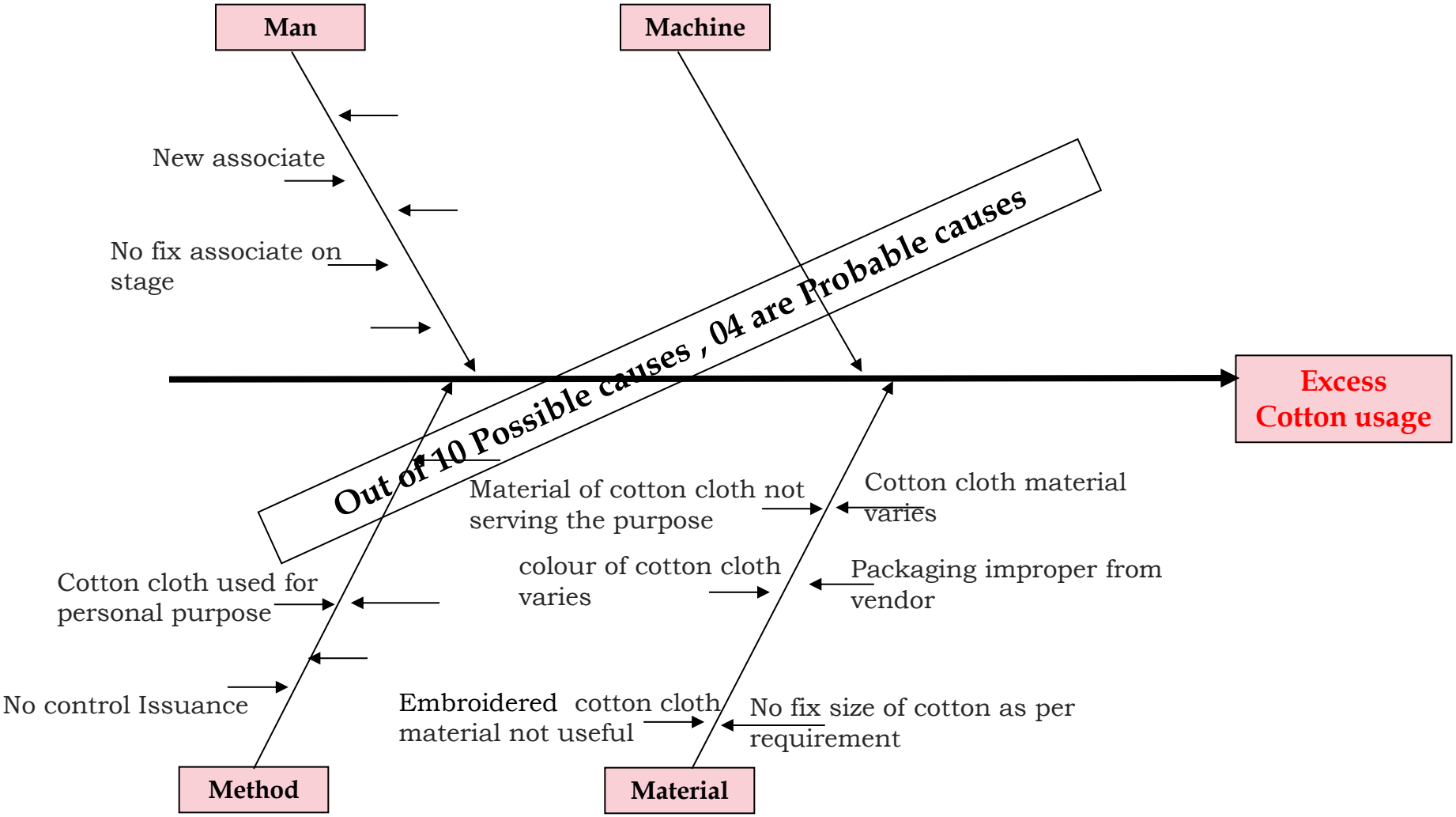
5. Analysis

List of Possible causes after brainstorming among team members

Sr. no	Possible Causes
1	New associate
2	No control Issuance
3	No fix size of cotton as per requirement
4	Cotton cloth material varies
5	Colour of cotton cloth varies
6	Cotton cloth use for personal purpose
7	Embroidered cotton cloth material not useful
8	Material of cotton not serving the purpose
9	No fix associate on stage.
10	Packaging improper from vendor.

Step 5. Analysis

5.1 Identify possible causes: Fish Bone diagram



Step 5. Analysis

List of Possible causes identification in Brain storming session

Sr. no	Possible Causes	Pravin	Nitin	Niranjan	Mahesh	Sonawane	Total
1	New associate	1	1	3	3	3	11
2	No control Issuance	9	3	3	9	3	27
3	No fix size of cotton as per requirement	9	9	9	9	3	39
4	Cotton material vary	3	3	1	1	3	11
5	color of cotton vary	3	9	9	9	9	39
6	Cotton use for personal purpose	9	3	1	1	3	17
7	designery cotton material not usefull	3	9	3	9	9	33
8	Material of cotton not surving the pupose	3	9	9	9	9	39
9	No Fix associate for usage of cotton	3	3	3	1	3	13
10	Packaging improper from vendor	3	3	1	3	3	13

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting cause for hypothesis is 36.

Step 5. Analysis

List of causes not considered as probable causes and reasons for not considering

Sr.No	Not considered Causes	
1	New associate	All trained associates deployed on stage
2	No control Issuance	Control issuance done , kept in locker
4	Cotton material vary	not a cause
6	Cotton use for personal purpose	Check Found ok
9	No Fix associate for usage of cotton	All trained associates deployed on stage
10	Packaging improper from vendor	Check Found ok as per PDS

Step 5. Analysis

List of causes considered as possible causes taken for test of hypothesis.

Sr. no	Possible Causes
3	No fix size of cotton as per requirement
5	color of cotton vary
7	designery cotton material not usefull
8	Material of cotton not surving the pupose

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
3	No fix size of cotton cloth as per requirement	By measuring tape.	900 sq cm cotton supplied by supplier, resulting to wastage of cotton.	**Valid**

Bigger cotton size after usage in paint booth , leads to wastage of cotton.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

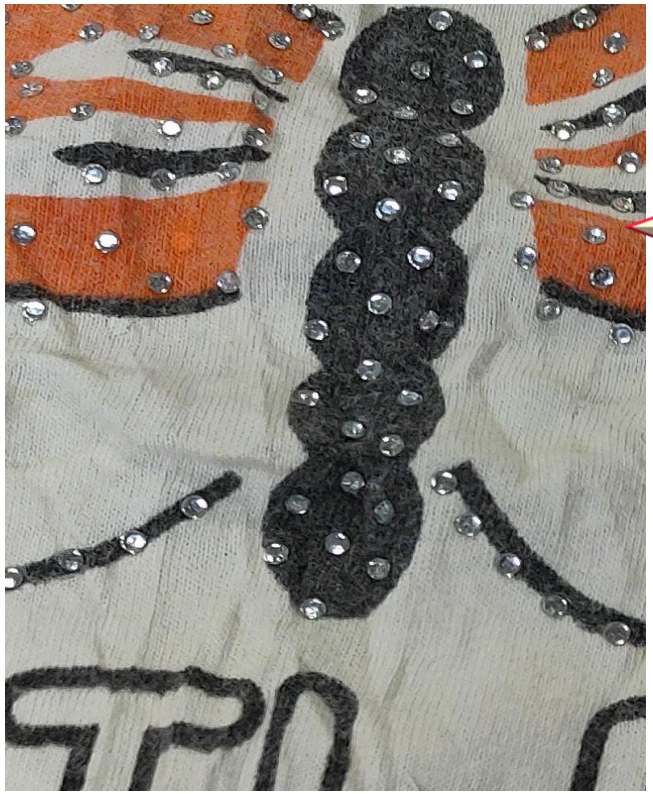
No	Probable Cause	Testing method	Testing and Observations	Conclusion
5	Color of cotton cloth Varyies	Visual	By cleaning vehicles by color full cotton cloth, it creates color stains on body .	**Valid**

Usage of colour full cotton creates stains on body leading to paint detoriation.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
	designery cotton cloth material not usefull	Visual	Usage of designer cotton cloth for cleaning & in paint booth creates scratches on Body of vehicle .	valid



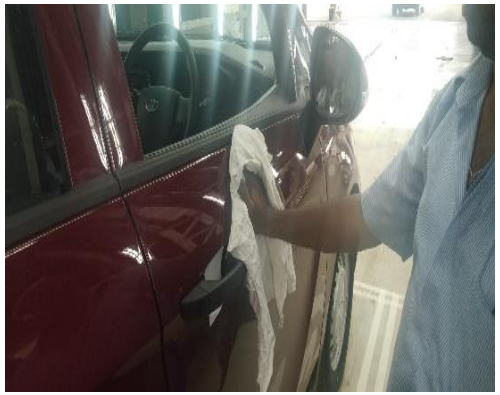
Usage of designer cotton for cleaning & in paint booth creates scratches on Body of vehicle .

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
	Material of cotton not serving the purpose for wiping.	Touch & feel	During vehicle cleaning cotton cloth does not soak the water, leading to high consumption.	**Valid**

During vehicle cleaning cotton cloth does not soak the water, leading to high consumption.



TEST OF HYPOTHESIS

5.3 Identify the root causes:

Sr No.	Probable cause	WHY?	WHY?	Action
1.	No fix size of cotton cloth as per requirement	No specific requirement for cloth size	Cloth used for various random purposes.	**Cloth size approximately defined.**
2.	Color of cotton cloth varies	Random cloth pieces of various material supplied by supplier.	No specific color requirement given to supplier.	White color cloth defined.
3	Designer cotton cloth material not useful	Generates scratches on body during usage.	Due to roughness & designer work	Order given for only plain white cloth
4	Material of cotton cloth not serving the purpose for wiping.	General purpose cloth , not soaking water completely from vehicle body during wiping	It is general cotton cloth, not made for soaking water.	Micro fibre cloth to be used.

6. Implement the corrective measures

Possible Remedies Identified

Sr. No.	Root Cause	Remedy Developed	Time to implement	Cost of Improvement	Easily reversible	Complexity	Decision
1	No fix size of cotton cloth as per requirement	**Required size to be define and supply to be ensured.**	Low	Low	Yes	Low	**Accepted**
2	Color of cotton cloth varies	White color cloth required.	Low	Low	yes	Low	**Accepted**
3	Designer cotton cloth material not useful	Plain cotton with no designer and diamonds to be supplied.	Low	Low	Yes	Low	**Accepted**
4	Material of cotton not serving the purpose for wiping.	For wiping and cleaning microfiber based cloth to be used.	High	High	Yes	Low	**Accepted**

6. Implement the corrective measures

6.1 Actions Decided

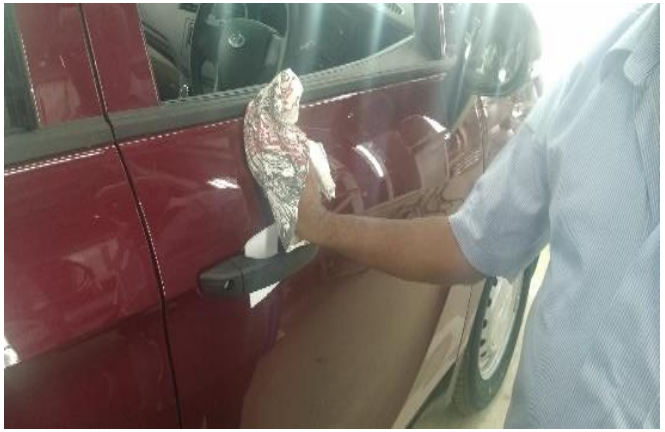
Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
1	No fix size of cotton cloth as per requirement.	Required size to be defined and supply to be ensured.	3.7.19	Nitin Aher	Closed

BEFORE	AFTER
900 Sq.cm. Cotton size approx. supplied by supplier for paint usage.	Reduced size 250 cm sq supplied by supplier.

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
2	Color of cotton Vary	White color cloth required.	6.8.19	Nitin Aher	Closed

BEFORE	AFTER
	
Colour cloth creates stain marks on body .	White colour cloth to be used.

6. Implement the corrective measures

6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
3	designery cotton material not usefull	Plain cotton with no designer and diamonds to be supplied.	6.8.19	Nitin AHer	Closed

BEFORE	AFTER
Usage of designer cotton generates scratches on body.	Plain cotton to be used .

6. Implement the corrective measures

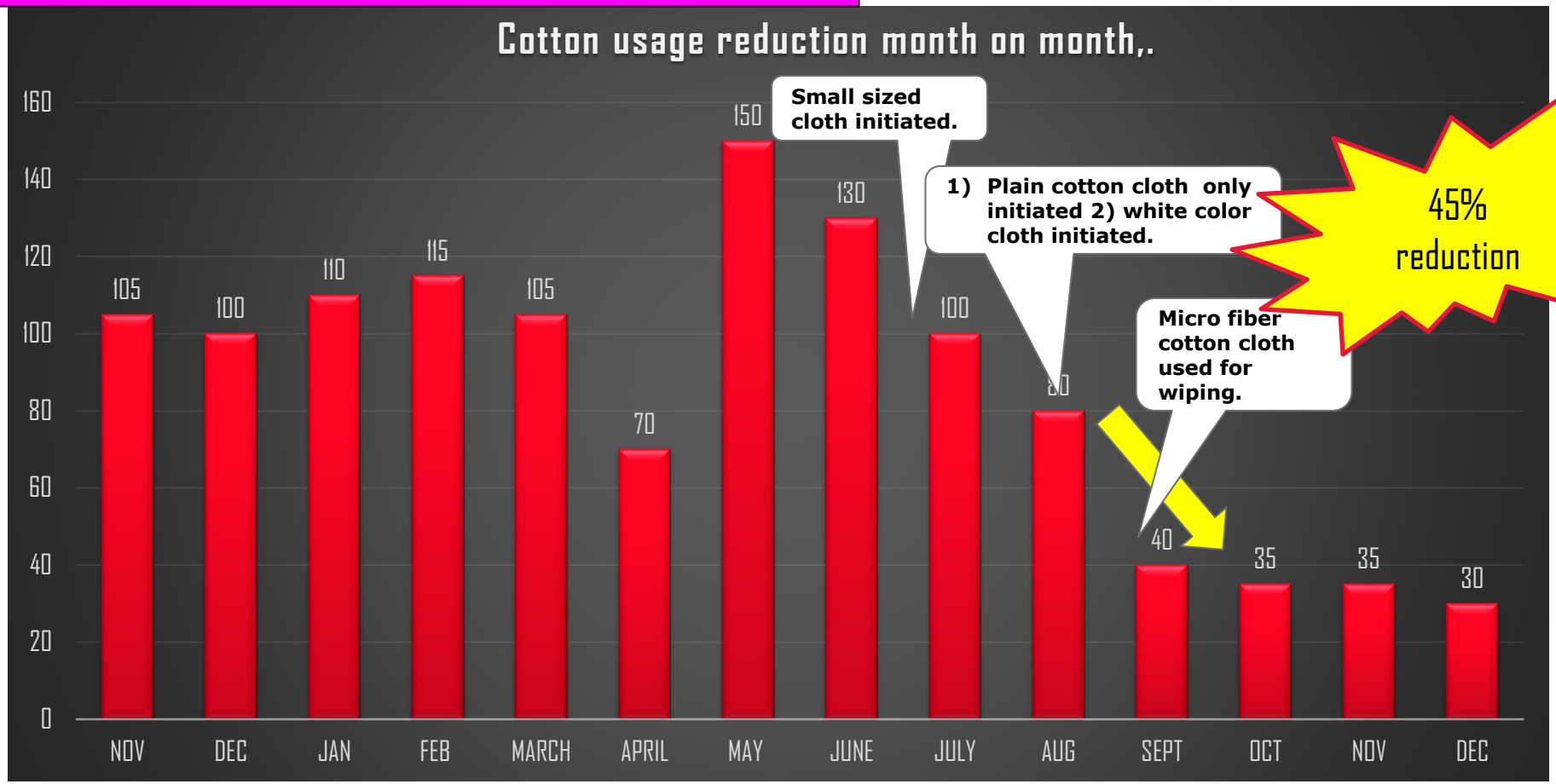
6.1 Actions Decided

Sr. No.	Root Cause	Action plan	Target Date	Responsibility	Status
4	Normal cotton not serving purpose .	Microfibre based cotton to be used.	4.9.19	Nitin Aher .	Closed

BEFORE	AFTER
Cotton cloth not serving the purpose of wet vehicle cleaning	Microfibre cloth to be used, which is fine fibre 300 times smaller than conventional cloth fibre, forces thousands of fibre points deep into the surface line with minimum effort resulting cleaning effectiveness.

7. Check & confirm the results & effects

7.1 Verify Results:



INFERENCE :- **As per graph usage of cotton reduced month on month after corrective actions.**

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Actual date	Responsibility	Status
1	Cotton cloth size reduced	03/07/19	Nitin Aher	Closed.
2	White color cotton cloth started.	6/8/2019	Nitin Aher	Closed.
3	Plain colour cotton cloth started.	6/8/2019	Nitin Aher	Closed
4	Microfibre based cotton cloth started.	4/9/2019	Nitin Aher	Closed

Check & confirm the results & effects (Cont.....)

7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) 45 % reduction in consumption of cloth.
2) RS 30160 Saved w.r.t F19
3) Reduction in hazardous cotton waste.

Intangible Benefits :

1) Learning of CFT working approach
2) Gemba Culture is learnt
3) Associate Morale Increased

From: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Sent: Tuesday, March 10, 2020 5:44:56 PM
To: AHER NITIN <AHER.NITIN@mahindra.com>
Cc: JOSHI SANDIP <JOSHI.SANDIP@mahindra.com>; BHAMARE SACHIN <BHAMARE.SACHIN@mahindra.com>; JADHAV NIRANJAN--NASHIK <JADHAV.NIRANJAN2@mahindra.com>
Subject: RE: Banyan cloth consumption

Okay Approved
This is related to CAI only

From: AHER NITIN
Sent: Tuesday, March 10, 2020 5:33 PM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Cc: JOSHI SANDIP <JOSHI.SANDIP@mahindra.com>; BHAMARE SACHIN <BHAMARE.SACHIN@mahindra.com>; JADHAV NIRANJAN--NASHIK <JADHAV.NIRANJAN2@mahindra.com>
Subject: FW: Banyan cloth consumption

Dear Rajesh,
Following is the consumption of Banian cloth in F19 and F20
Part no: 4903022
In F20 we achieved reduction of 46% against F19
Pl approve , Details as follows
TCF

Month	F19	F20	Diff in Kgs	Reduction in kgs	Reduction %
Apr	100	70	-30	70	30
May	110	150	40	136	-36
Jun	90	50	-40	56	44
Jul	175	100	-75	57	43
Aug	215	70	-145	33	67
Sep	150	105	-45	70	30
Oct	160	50	-110	31	69
Nov	115	30	-85	26	74
Dec	100	35	-65	35	65
Total	1215	660	-555	54	46

Regards,
Nitin Aher
STCF

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Actual date	Responsibility	Status
1	Cotton cloth size reduced	03/07/19	NITIN AHER	Closed
2	White color cotton cloth started.	6/8/2019	NITIN AHER	Closed
3	Plain white colour cotton cloth started.	6/8/2019	NITIN AHER	Closed
4	Microfibre based cotton cloth started.	4/9/2019	NITIN AHER	Closed

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Recycle of cloth – Training Given .	10/7/2019	Nitin Aher	Closed

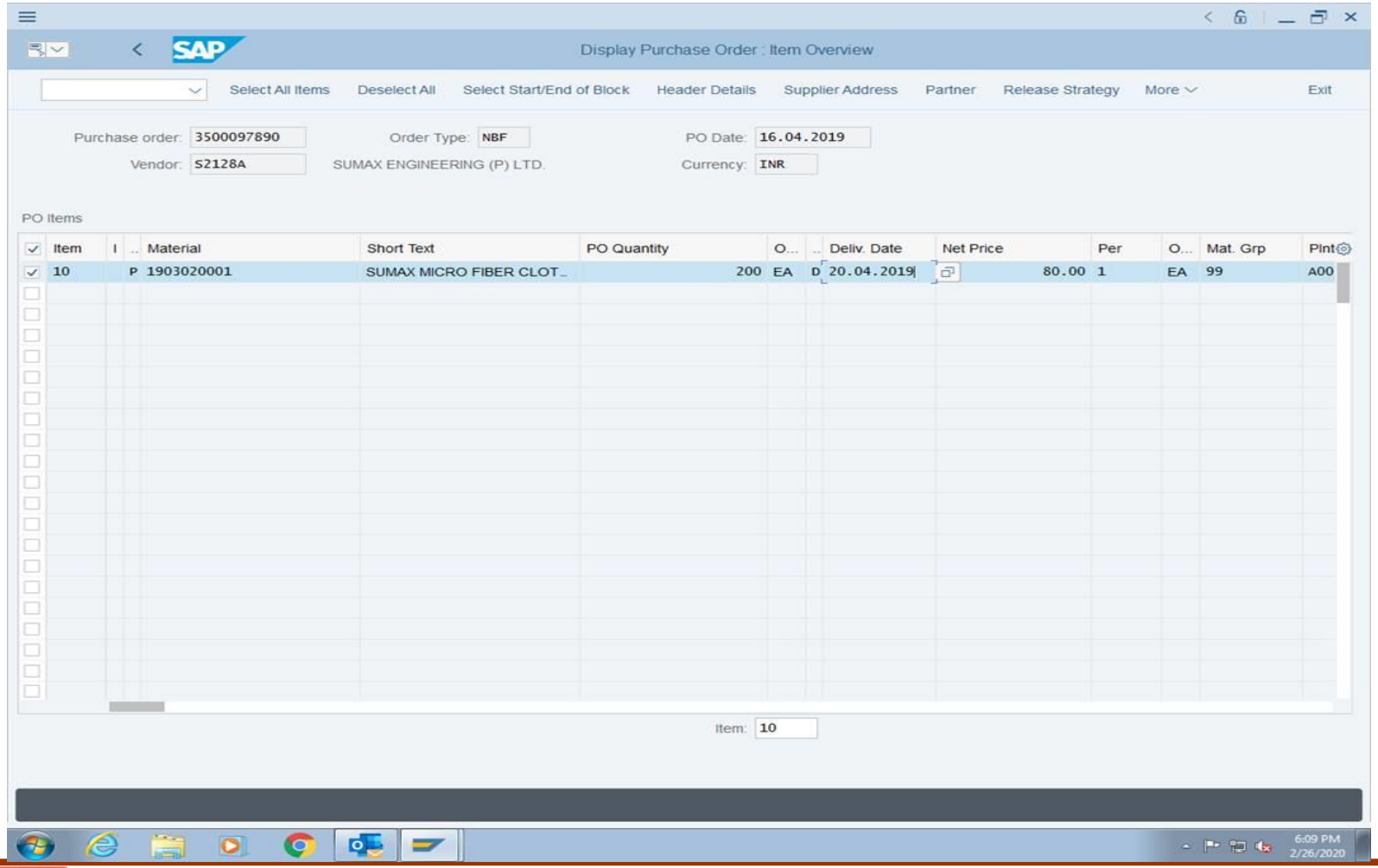
8.3 Additional Audit & check

	NA			

8. Standardization

8.1 Updation of documents:

PO FOR NEW MICROFIBRE COTTON



8. Standardization

8.2 Training:

SN	Training Identified	Actual date	Responsibility	Status
1.	Recycle of cloth – Training Given .	10/7/2019	Nitin Aher	Closed

9. Summarize & Plan future actions

9.1 Conclusion:

By close monitoring & brain storming usage of FOS can be reduce with same outputs.

9.2 Horizontal Deployment:

Analysis is useful in upcoming projects, inputs to DFMEA.

9.3 Lessons Learnt:

Things Gone Wrong :

Monitoring not done for FOS consumption.

Things Gone Right :

Brainstorming done between CFT team & problem solved by QC story approach.

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Thank you

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